



LectureAssist: An Agentic, Multimodal AI Assistant for Lecture-Grounded Study Material

Iftikhar Ahmed · Salman Shafi · Jarinah Tasnim
Supervisor: Dr. Adnan Areefen

Problem

- Lecture knowledge is locked in **video, speech and board writing**.
- Generic quiz tools produce **ungrounded, off-topic** questions.
- Fixed difficulty ignores **each student's weak topics**.

Goal. Grounded, validated, adaptive study material — fully local, **zero API cost**, on one free Colab T4.

Evaluation Setup

- Source.** OpenStax *Calculus Vol. 1*, Limits chapter ($\approx 112k$ characters), also given to the judge as ground truth.
- Scale.** $n=200$ MCQ per pipeline — easy 68, medium 66, hard 66.
- Fairness.** Identical sliding window for both pipelines, so any difference reflects **prompting strategy alone**.
- Models.** Generator **gemma3:4b**, judge **gemma3:12b**, both local.

Stack

- Vision.** EasyOCR, per-surface preprocessing for slides, chalkboards and whiteboards.
- Speech.** Faster-Whisper with per-segment timestamps.
- Retrieval.** MiniLM embeddings in a FAISS cosine index.
- Serving.** Flask on Colab $\rightarrow$ ngrok $\rightarrow$ browser UI.

End-to-End Flow

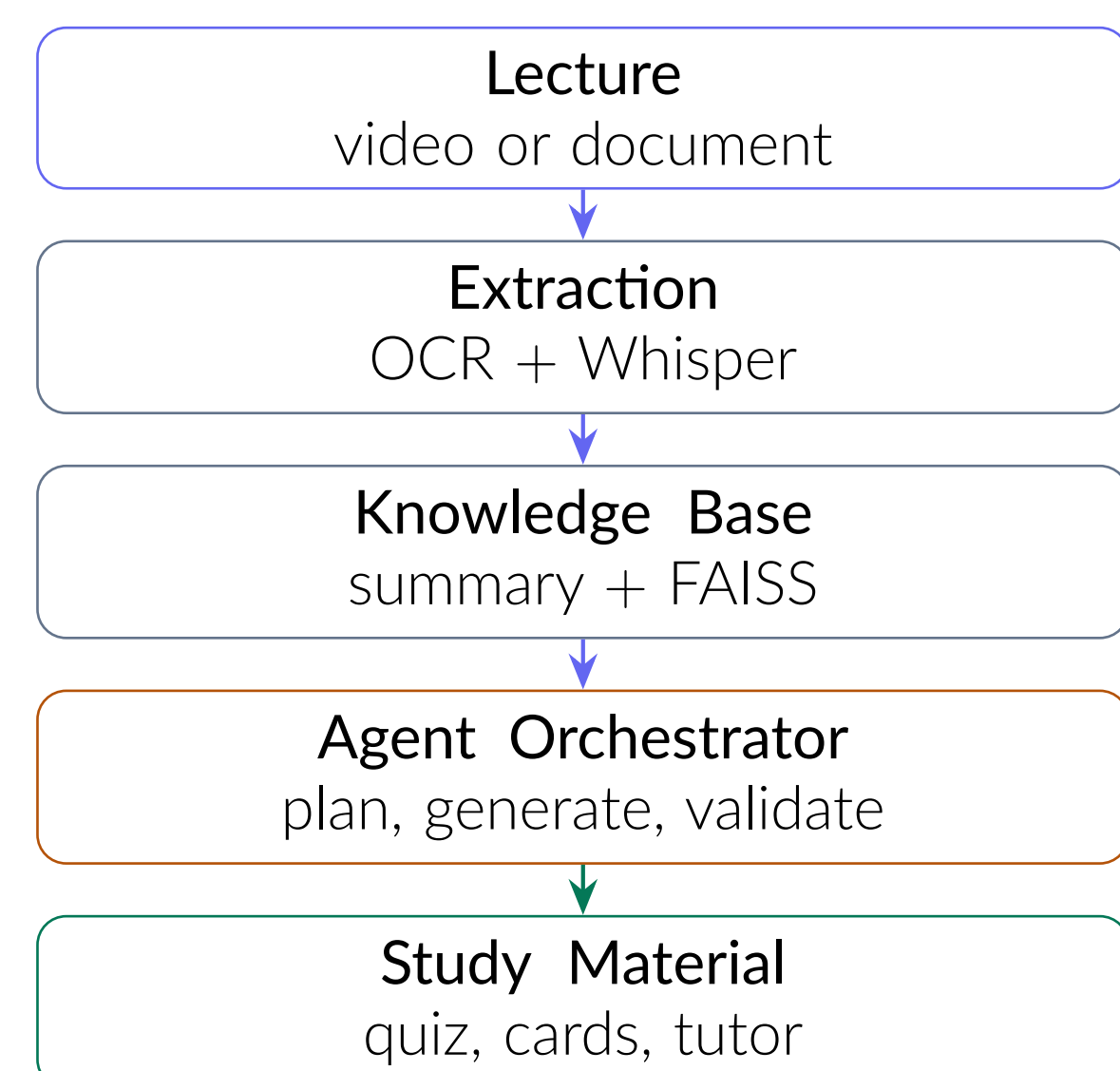


Figure 1. Everything runs locally on one free Colab T4.

System Architecture

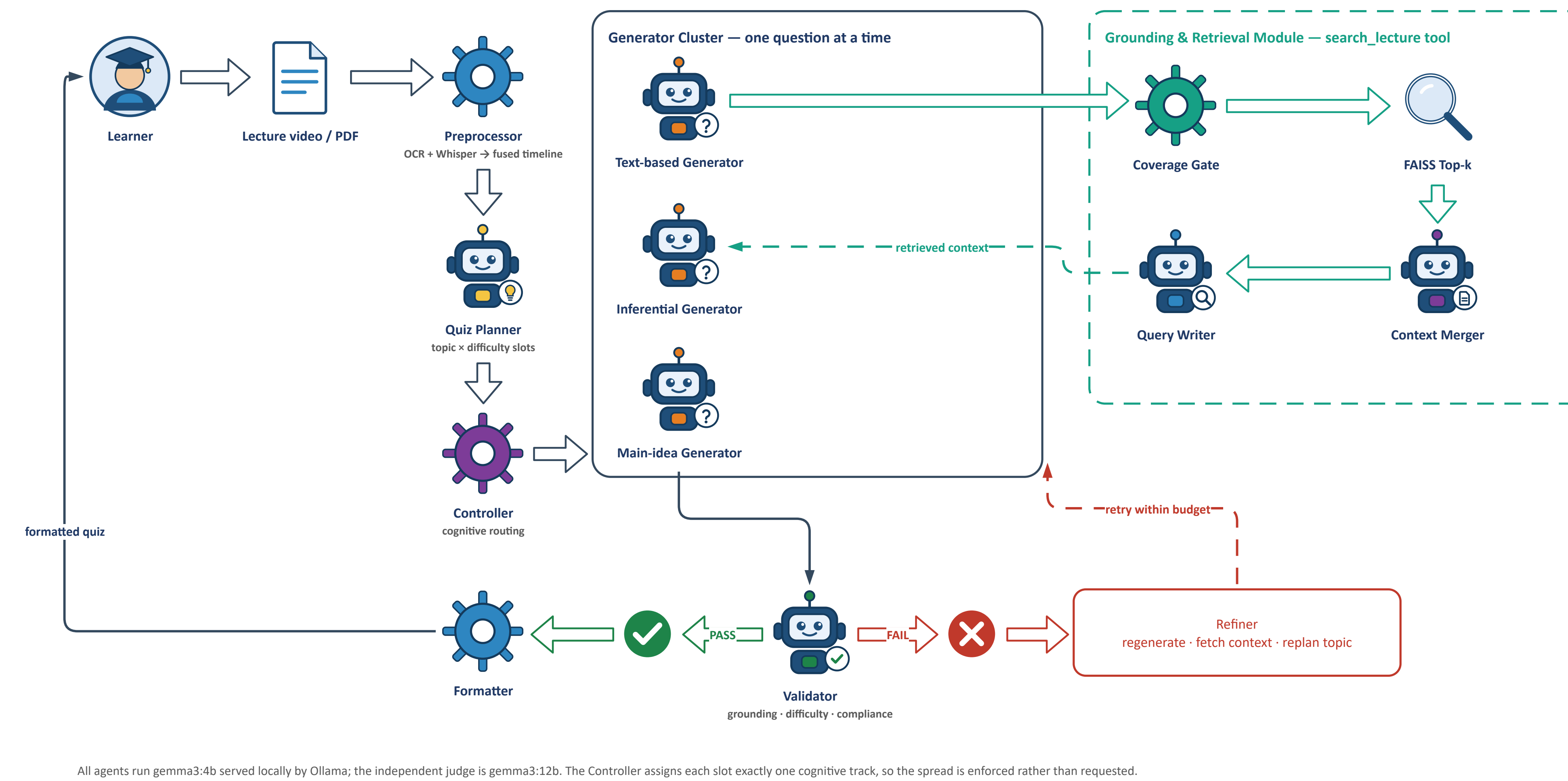


Figure 2. Multimodal extraction feeds a grounded knowledge base; the orchestrator coordinates the specialist agents that produce every study artifact. Independent agents run concurrently and any agent crash is isolated.

Agentic Quiz Generation

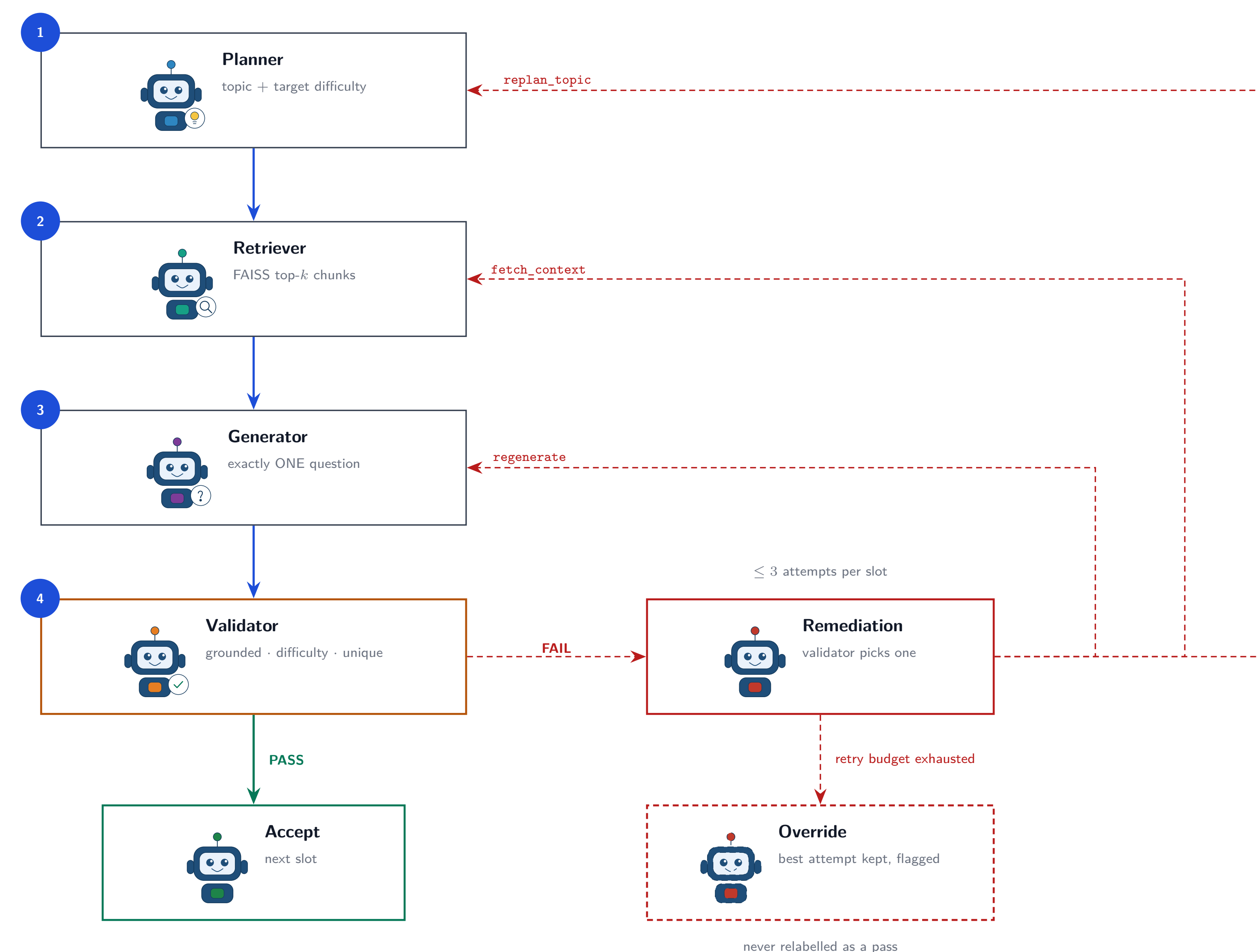


Figure 3. One question per slot; each must pass validation before the next begins. An exhausted retry budget flags an override, never a pass.

Results

Metric	Agen.	Non-ag.
<i>Judged 1–5 (saturated)</i>		
Overall	4.86	4.93
Correctness	4.98	5.00
Clarity	4.92	4.97
Difficulty	4.36	4.39
<i>Automated 0–1</i>		
Duplicate ↓	0.385	0.280
Grounding	0.557	0.624
Eff. accept	0.612	0.720
Answer match	0.940	0.970
Similarity ↓	0.932	0.894
Accept rate	0.995	1.000
Verified	0.995	0.995

$n=200$ per pipeline. Green marks the better value; ↓ marks rows where lower is better. Per-difficulty $n=68/66/66$.

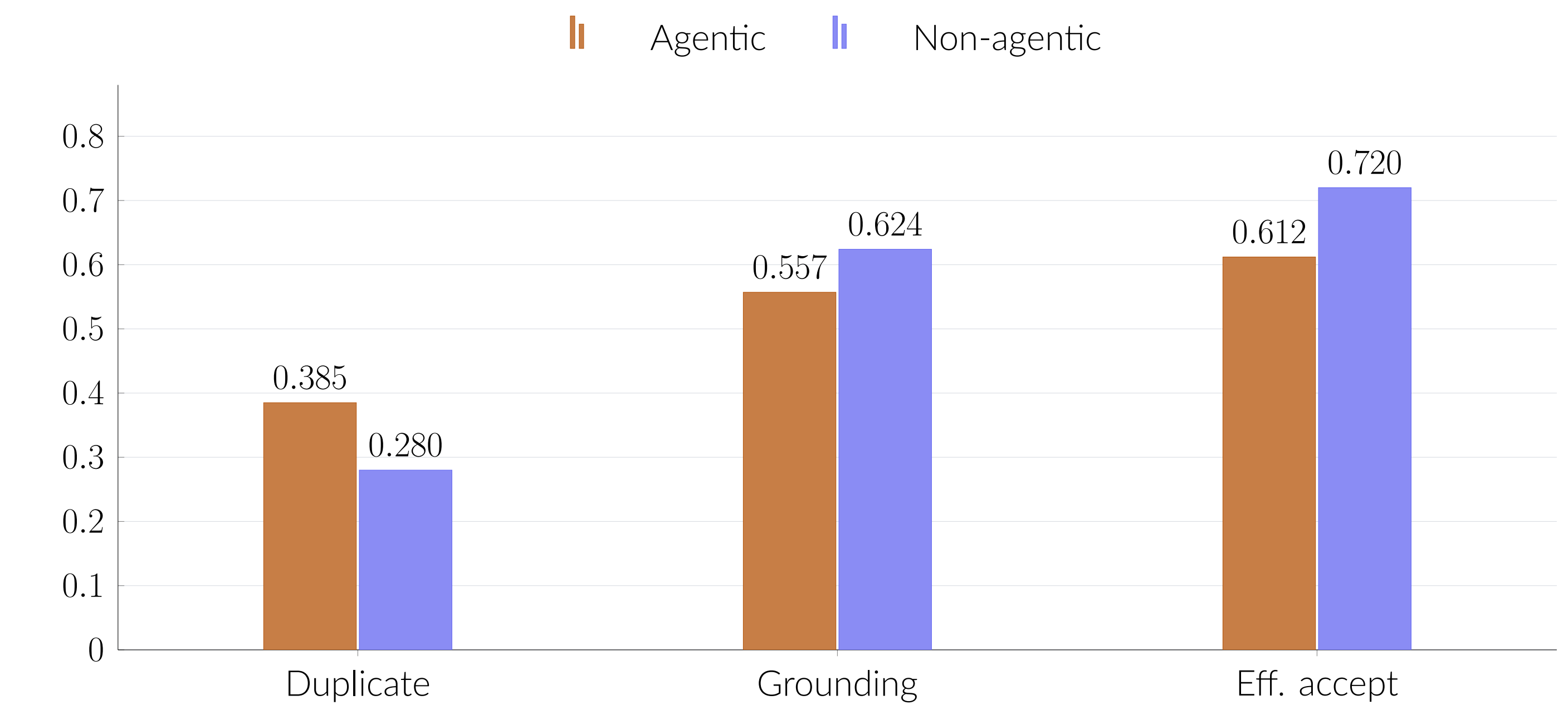


Figure 4. The three metrics that discriminate. Lower is better for duplicate rate, so the baseline leads on all three.

Conclusion

A fully local, \$0-API multimodal pipeline that turns a lecture into grounded, validated study material. The judged dimensions **saturate** (4.36–5.00), so only the automated metrics discriminate — and on those the agentic loop does **not** beat single-pass prompting. The engineering contribution stands; the scientific claim is **not established**, and we report it that way.

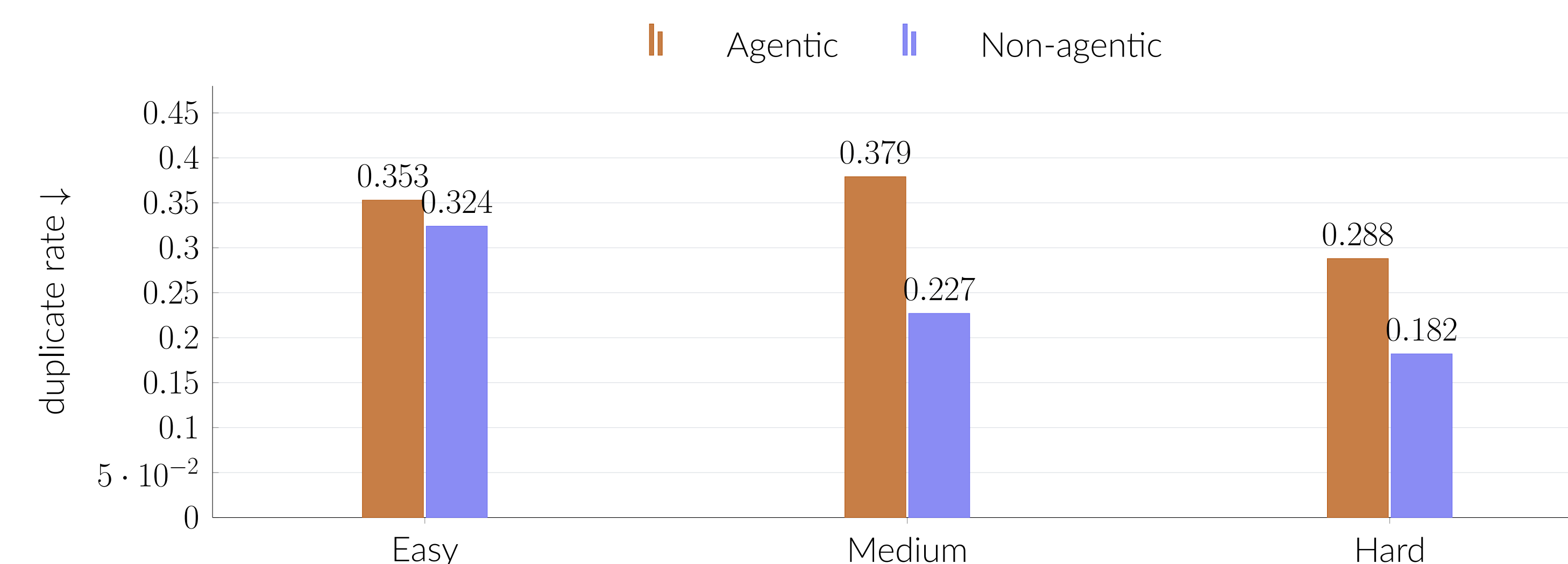


Figure 5. Duplicate rate by difficulty — the agentic loop duplicates more in every band.